

Science · GROW · Autumn 1 · Week 08 · SCI-GROW-A1-W08-P1 · Date _____

The Moon's JourneyToday I can... ☐ State that the Moon orbits Earth in roughly a month ☐ Explain that moonlight is reflected sunlight ☐ Explain phases using our changing view of the lit halfFeedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R _____

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**The Moon's Journey · Route card**

Reference only · record your answers in your book.

Independent

Sequence four model positions and explain why the visible lit part changes.

Supported

Match new, quarter, full and quarter views with a partner. Use a fixed sunlight arrow and point to the lit half.

Choice

Use phase cards to label four model views. Choose each label: full / quarter / new. Order the labelled views around the model. Mark sunlight direction and the lit half. Choose the cause: our changing view of the lit half / the Moon changing shape. Complete: We see ____ because _____. Choose the lunar eclipse cause: Earth blocks sunlight / the Moon becomes smaller. Record C beside your book response.

Word bank

- satellite: an object that orbits a planet
- phase: the shape of the lit part of the Moon visible from Earth
- orbit: a curved path around another body
- reflected: light that changes direction from a surface

Model steps

- Torch is the Sun, ball is the Moon, observer is Earth. Keep the sunlight direction fixed.
- Half the Moon is illuminated by the Sun. As it orbits, our view of the illuminated half changes.
- When the visible lit half faces us, we see full Moon. When it faces away, we see new Moon.

Kit: Cool LED torch, large foam ball, observer marker and four phase cards.

Safe: Adult uses a low-power cool light; no lasers or looking at the Sun. No blackout: normal-light diagrams are equal routes. Keep the ball above the observer's head shadow so a false eclipse is not created. Agree before any light change; adult handles equipment and cutting. Keep movement clear.

Aspire

Explain the phase sequence, distinguish eclipses and name one model limit.

Separate changing viewpoint from blocked sunlight; explain the tilted orbit.

Staff mark I, S or C beside this task.

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**The Moon's Journey**

Reference only · record your answers in your book.

Learning goal: I can explain why the Moon appears to change shape as it orbits Earth.

Word	Meaning
satellite	an object that orbits a planet
phase	the shape of the lit part of the Moon visible from Earth
orbit	a curved path around another body
reflected	light that changes direction from a surface

Key ideas

- Torch is the Sun, ball is the Moon, observer is Earth. Keep the sunlight direction fixed.
- Half the Moon is illuminated by the Sun. As it orbits, our view of the illuminated half changes.
- When the visible lit half faces us, we see full Moon. When it faces away, we see new Moon.
- Trace the light path: Sun → Moon → observer on Earth. Name the Sun as light source and the Moon as reflecting object.

Keep the claim fair

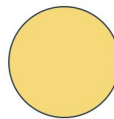
The model does not show realistic sizes, distances or the tilt of the Moon's orbit.



new Moon



half Moon



full Moon

Views from Earth; not to scale. The ball stays round at every position.

New, half and full are views from Earth. The Moon stays round; its visible lit part changes.

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**Staff: cut these apart before the lesson.**

Move these pieces for the task. Record the answer and reason in your book.

Crescent: Only a small part of the illuminated half faces us.	Full Moon: The illuminated side faces our observer.
New Moon: The lit half faces mostly away from the observer.	Dark red Moon: Earth blocks direct sunlight during this event.
Own light: The Moon glows by itself.	Every month: Every full Moon must be an eclipse.

Method strip

1. Keep the cool light fixed and use one Moon model or the prepared phase cards.
2. Mark the lit half and the observer position. Stop and check the light direction with the adult.
3. Sequence four views from the same observer, keeping phase and eclipse explanations separate.
4. In your book, label an observation and conclude why the visible lit part changes. Restore lighting and return equipment.

After the practical: record an observation and one conclusion linked to today's outcome in your book.